

Assignment 1: Computer Architecture, Organization, and Performance Measurement

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Deadline: two weeks from the date of issue — late submissions lose 10% per day unless a documented excuse is provided in advance (see the course policy in the syllabus).

Answer every question. Show all working for Numerical and Design questions — a bare final answer without the derivation earns partial credit only.

1.1 Conceptual

Q1. A new model of an existing x86 processor ships with a redesigned datapath that halves the CPI of most instructions but keeps the instruction set, registers, and addressing modes unchanged. A binary compiled years ago runs unchanged and measurably faster on the new model. Explain this observation using the distinction between *computer architecture* and *computer organization*: which of the two changed, which stayed fixed, and why old software continues to work. Your answer must use both consequences — compatibility and design freedom — from the lecture.

Q2. Two machines run the same program P . Machine A reports $\text{MIPS} = 1000$; Machine B reports $\text{MIPS} = 800$. A student concludes that A is faster. Explain why the MIPS numbers alone cannot support that conclusion, state the metric that *should* be used instead, and identify the one special condition under which the MIPS comparison would happen to be a safe guide.

1.2 Numerical

Q3. Program P executes 2×10^8 instructions. Machine A runs at $f = 2.5$ GHz with $\text{CPI} = 1.8$; Machine B runs at $f = 3.0$ GHz with $\text{CPI} = 2.2$. Compute the CPU time of both machines, state which is faster, and by what factor.

Q4. A program's total execution time is 200 s. A new floating-point unit makes the FP portion of the program $3\times$ faster. Compute the overall speedup, and separately compute the *best possible* overall speedup (i.e. the ceiling when the FP part is made infinitely fast).

1.3 Design

Q5. A program spends 25% of its time on load/store instructions. An engineer proposes a new memory system that makes loads/stores $8\times$ faster. (a) Compute the resulting overall speedup. (b) Compute the hard ceiling on overall speedup that this proposal can ever reach, no matter how much faster loads/stores become. (c) The engineering target is a $2\times$ overall speedup — using your

two numbers, argue whether investing further in the load/store path is the right place to spend design effort, and where the remaining gain must come from.

Q6. You are required to achieve a $4\times$ overall speedup on a workload by improving only its floating-point portion, which you may make *infinitely* fast. Using Amdahl's Law, determine the minimum fraction f of execution time that the FP portion must occupy for this goal to be achievable, and give the resulting speedup expression that confirms it.